nature energy

Article

https://doi.org/10.1038/s41560-024-01577-9

Improving the SO_2 tolerance of CO_2 reduction electrocatalysts using a polymer/catalyst/ionomer heterojunction design

Received: 25 October 2023

Accepted: 10 June 2024

Published online: 4 July 2024

Check for updates

Panagiotis Papangelakis[1,9], Rui Kai Miao[1,9], Ruihu Lu[2,9], Hanqi Liu[3,9], Xi Wang[3,4,9], Adnan Ozden[1], Shijie Liu[1], Ning Sun[3], Colin P. O'Brien[1], Yongfeng Hu[5], Mohsen Shakouri[6], Qunfeng Xiao[6], Mengsha Li[7], Behrooz Khatir[1], Jianan Erick Huang[8], Yakun Wang[8], Yurou Celine Xiao[1], Feng Li[1], Ali Shayesteh Zeraati[1], Qiang Zhang[3], Pengyu Liu[3], Kevin Golovin[1], Jane Y. Howe[7], Hongyan Liang[4], Ziyun Wang[2] ✉, Jun Li[3] ✉, Edward H. Sargent[8] & David Sinton[1] ✉

The high concentrations of CO_2 in industrial flue gases make these point sources attractive candidates for renewably powered electrocatalytic conversion of CO_2 to products. However, trace SO_2 in common flue gases rapidly and irreversibly poisons catalysts. Here we report that limiting hydrogen adsorption in the vicinity of electrochemically active sites deactivates SO_2 to enable efficient CO_2 conversion. We realize this approach via a polymer/catalyst/ionomer heterojunction design with combined hydrophobic and highly charged hydrophilic domains that diminish hydrogen adsorption and promote CO_2 over SO_2 transport. We develop an SO_2-tolerant system that maintains ~50% faradaic efficiency towards multi-carbon products for over 150 h (at 100 mA cm^{-2}). Extending this strategy to a high-surface-area composite catalyst, we achieve faradaic efficiencies of 84%, partial current densities of up to 790 mA cm^{-2} and energy efficiencies of ~25% towards multi-carbon products with a CO_2 stream containing 400 ppm SO_2, a performance that is competitive with the best reports using pure CO_2.

Powered by renewable electricity, electrochemical CO_2 reduction (CO_2R) into fuels and feedstocks offers a means to mitigate persistent greenhouse gas emissions[1]. To date, the field has focused on the electrocatalytic conversion of high-purity CO_2 (refs. 2–4). However, generating a high-purity CO_2 reactant stream—from air or industrial point sources—is costly, requiring as much or more energy than the electroconversion step[2,5]. Achieving viable electroproduction of valuable large-market products such as ethylene and ethanol will require lowering the energy costs associated with both the conversion process and the CO_2 reactant supply[2,6,7].

Direct air capture systems (Fig. 1) can produce high-purity CO_2 for subsequent CO_2R (refs. 8,9). Pilot-scale systems use alkali solutions or amine-based solid adsorbents to capture CO_2, which involves a series of drying and calcination (>900 °C) steps to regenerate the capture media and release concentrated CO_2 (refs. 10,11). This regeneration process requires 100–300 kJ mol^{-1} of thermal energy input and produces CO_2 emissions that are counterproductive to the overall process, particularly when the thermal energy is provided from fossil fuels[12]. Therefore, coupling direct air capture and CO_2R processes in series leads to high overall energy costs. The current cost of direct air capture is estimated to be US$600 tCO_2^{-1}, with the potential to be reduced to US$200 tCO_2^{-1} with substantial technological advancements[5]. However, a technoeconomic analysis of CO_2 electrolysers shows that, even at projected costs, the CO_2 capture step accounts for over half of the

A full list of affiliations appears at the end of the paper. ✉e-mail: ziyun.wang@auckland.ac.nz; lijun001@sjtu.edu.cn; sinton@mie.utoronto.ca

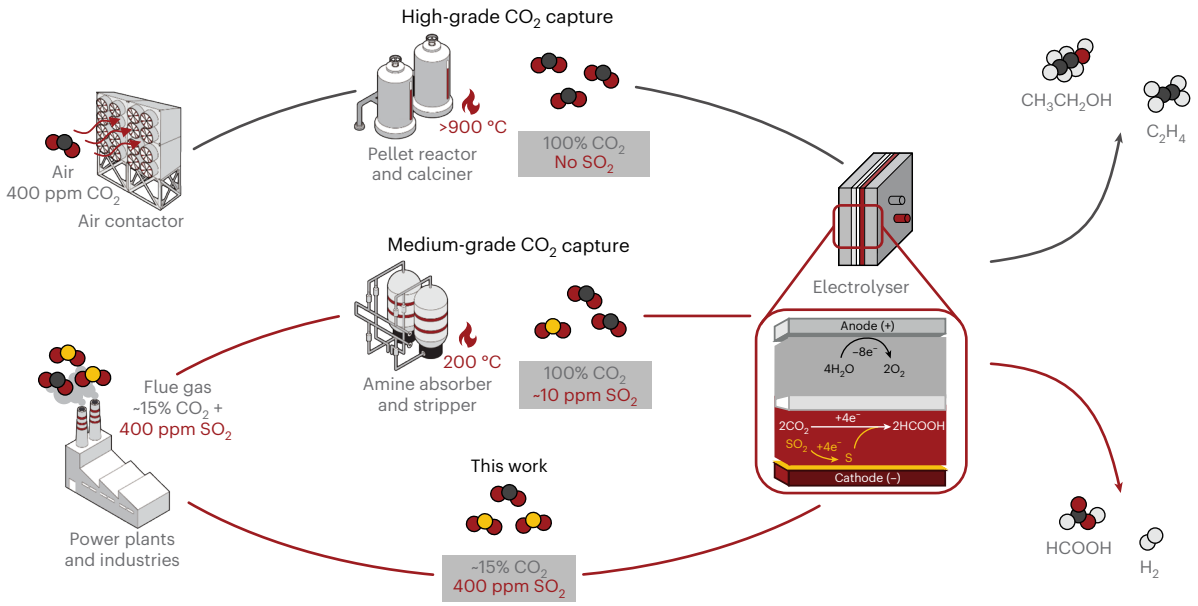

Fig. 1 | Processing of CO_2 gas. The capture and electrolysis of CO_2 from air or flue gas streams and the effect of SO_2 poisoning on the reaction. e^-, electron.

total operating costs (Supplementary Note 1, Supplementary Fig. 1 and Supplementary Tables 1 and 2).

Employing CO_2 capture liquid as a reactant stream is one approach to circumvent the CO_2 purification step. In this reactive capture approach, carbonate or carbamate solutions are the direct input to the electrolyser. However, this method has generated mostly single-carbon (C_1) products, such as CO, formate and methane, and has been limited to low current densities[2,7,13–22], with one recent exception of multi-carbon (C_{2+}) product generation at a partial current density of 141 mA cm^{-2}. The low productivity and low selectivity of the current reactive capture approaches result in high capital and operating costs[23]. As a result, prices for products made using atmospheric CO_2—via reactive capture or sequential capture and conversion—far exceed market prices[1,5,24]. Therefore, as the performance of CO_2R systems continues to improve[25–27], the relative cost of the capture step grows, motivating an alternative approach.

Industrial CO_2 point sources can provide CO_2 at concentrations 100 times that of air, and these emissions are substantial, growing in intensity and some of the most challenging to abate. Effluent streams from industrial sources account for 50–60% of anthropogenic CO_2 emissions and offer an avenue to eliminate CO_2 capture and purification costs (Fig. 1)[28]. Flue gas sources have CO_2 concentrations varying between 10 and 60%—a range that can be productively reacted in suitably designed CO_2R systems[29,30]. Despite these advantages, the direct use of flue gas in CO_2 for electrolysis has not been achieved due to the negative effects of gas impurities[8,31]. Chief among these is sulfur dioxide (SO_2), which has a typical concentration of ~400 ppm in non-coal-fired flue gas streams and can still persist at ~10 ppm after gas scrubbing[32–34]. Even low concentrations of SO_2 have been shown to rapidly and irreversibly poison CO_2R catalysts, resulting in a complete loss of C_{2+} selectivity within 20 min of operation[8,31].

Herein we aimed to elucidate the reasons behind the low C_{2+} selectivity observed when co-feeding SO_2/CO_2 for CO_2R and develop a catalyst resistant to SO_2 poisoning. Our investigation revealed that SO_2 readily undergoes hydrogenation with adsorbed hydrogen (*H) on the catalyst, leading to the accumulation of *H and *S species on the catalyst surface. This accumulation hinders the active sites, impedes CO_2 adsorption and thereby triggers the hydrogen evolution reaction (HER) and a decline in C_{2+} selectivity. We also report an increase in the formation of formate from the reaction of CO_2 with *H instead of solvated H, resulting in decreased CO formation and reduced CO dimerization—a crucial step in the production of C_{2+} products. Thus, the concurrent build-up of *H and *S species on the catalyst due to SO_2 reduction reduces C_{2+} selectivity in the CO_2R electrolyser. Motivated by these findings, we designed a polymer/catalyst/ionomer heterojunction that integrates hydrophobic and highly charged hydrophilic domains to address the SO_2 poisoning effect. This approach aims to curb hydrogen adsorption and enhance the transport of CO_2 over SO_2. The hydrophobic components facilitate the unobstructed passage of non-polar CO_2 and limit the absorption of hydrogen, while the hydrophilic domains exhibit strong interactions with polar SO_2, impeding its transport. As a result, we achieved the electrosynthesis of C_{2+} products with faradaic efficiency (FE) values up to 84% and partial current densities reaching up to 790 mA cm^{-2} with a CO_2 stream containing 400 ppm SO_2.

Analysis of the SO_2 poisoning mechanism

Initial tests using a dilute CO_2 stream (50%, balanced with N_2) containing trace SO_2 showed that a low concentration of 10 ppm SO_2 degraded the FE of C_{2+} products (FE_{C2+}) from 64% to 36% within two hours of electrolysis (Fig. 2a and Supplementary Fig. 2; 200 nm sputtered copper catalyst at 100 mA cm^{-2} using a membrane electrode assembly[35]). At the same electrolysis conditions, an increase in SO_2 concentration up to 400 ppm (typical of raw flue gas) resulted in a rapid loss of FE_{C2+} to 23% after only 10 min and a further decrease to 11% over two hours of operation. In both cases, hydrogen and formate production became dominant. The reduction of SO_2 is expected to form persistent sulfur deposits on the catalyst, resulting in an increased cell resistance and cell potential (Supplementary Fig. 3)[36]. These results are akin to previous reports on CO_2R in which acute changes in product selectivities from C_{2+} products to hydrogen and formate were observed with Cu catalysts in the presence of trace SO_2 (refs. 8,31). The rapid and irreversible poisoning of Cu in the presence of SO_2—even at low, post-scrubber concentrations—indicates that current CO_2R approaches are incompatible with flue gas sources of CO_2 and thus inapplicable to over half of anthropogenic emissions[8].

We took the view that the increase in HER associated with SO_2 exposure indicates that the hydrogenation of the SO_2 molecules on the catalyst surface is central to the poisoning effect. Given the greatly increased surface wettability of the catalysts following SO_2 exposure (Fig. 2b), we posited that an accumulation of adsorbed hydrogen (*H) causes a loss of CO_2R selectivity upon SO_2 poisoning. This

https://doi.org/10.1038/s41560-024-01577-9

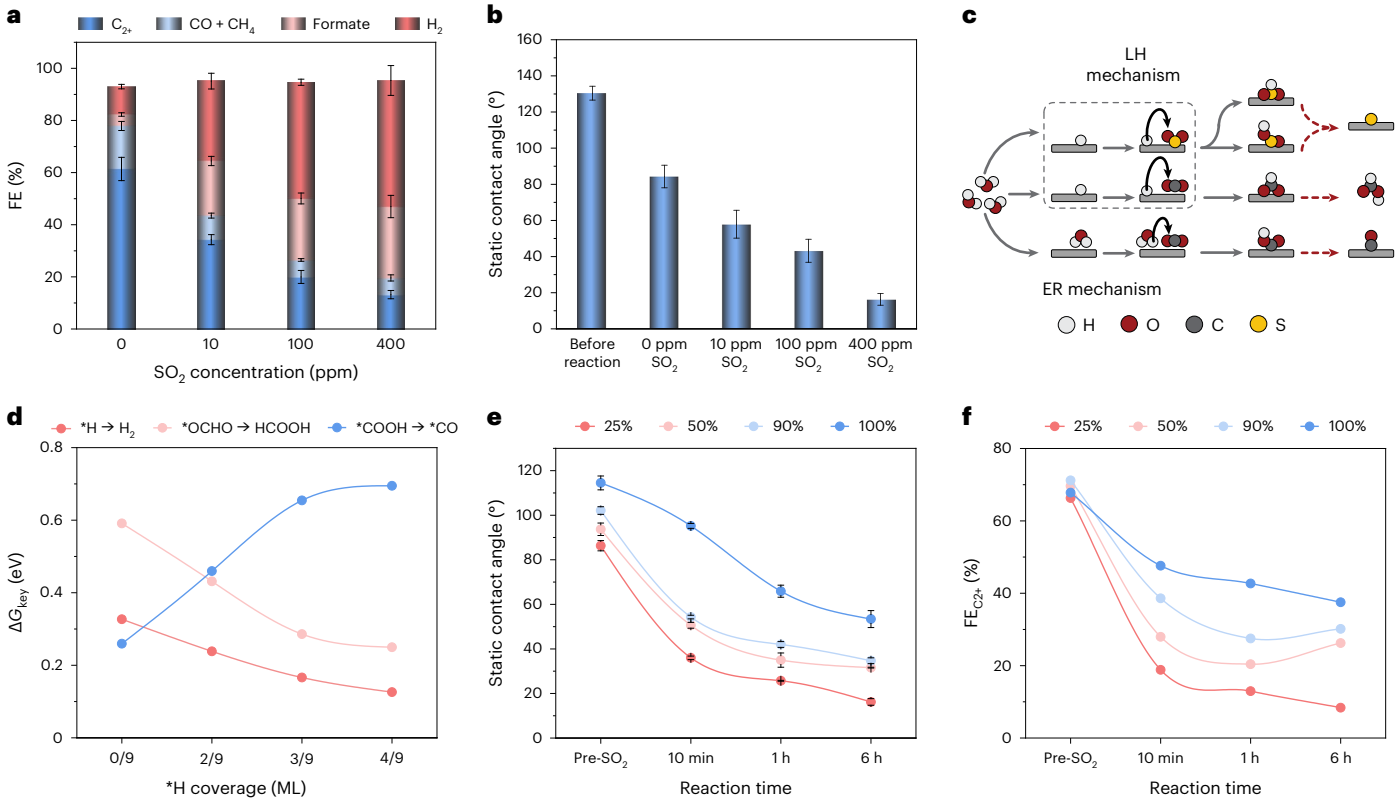

Fig. 2 | Investigation of the SO_2 poisoning mechanism. a, Distribution of product selectivities from the electrolysis of dilute CO_2 (50%, balanced with N_2) at various SO_2 concentrations in a standard membrane electrode assembly electrolyser. The cell was operated at a constant current density of 100 mA cm^{-2}. Values are means, and error bars indicate s.d. ($n = 3$ replicates). **b**, Water contact angle measurements on a 90%-PTFE-supported Cu electrode (200 nm sputtered Cu). Measurements were taken both before and after 6 h of CO_2R reaction with 0–400 ppm SO_2 as shown at a constant current density of 100 mA cm^{-2}. Values are means, and error bars indicate s.d. ($n = 4$ measurements). **c**, Proposed hydrogenation mechanisms for SO_2 activation towards S formation and CO_2 reduction towards formate formation and CO formation. **d**, The change of free energies of the key elementary steps with respect to *COOH formation (for *CO production) and desorption of HCOOH and H_2, respectively, at various *H coverages. **e**,**f**, The change of static contact angles (**e**) and C_{2+} product selectivities (**f**) of Cu electrodes supported by various PTFE loadings at different reaction times from the co-electrolysis of CO_2 and SO_2. Experiments were performed at a constant current density of 100 mA cm^{-2} using a gas feed comprising dilute CO_2 (50%, balanced with N_2) and 400 ppm SO_2. Where error bars are shown, values are means and error bars indicate s.d. ($n = 4$ measurements).

adsorbed hydrogen steers the CO_2 hydrogenation pathway from an Eley–Rideal (ER)-type reaction using a solvent proton source (that is, water) to a Langmuir–Hinshelwood (LH)-type reaction with adsorbed H (Fig. 2c), inhibiting the production of *CO, a key intermediate for C_{2+} production[37].

Theoretical calculations

We began by testing our hypothesis on a Cu surface using density functional theory (DFT) calculations (details in Supplementary Note 2), assessing the effect of the proton source (that is, adsorbed H or water) on the thermodynamics and kinetics of the first hydrogenation reactions of SO_2/CO_2. To do so, we constructed adsorbed H species (*H) on a Cu surface and solvent proton species in a charged water layer[38] on Cu to represent the surface and solvent proton sources, respectively. The DFT results show that the hydrogenations of SO_2/CO_2 with adsorbed H form *SOOH, *HSO_2 and *OCHO intermediates that are energetically favoured over those formed with water; however, solvent protons are more active than adsorbed H in lowering the energy barriers of CO_2 hydrogenation to produce *COOH (Supplementary Figs. 4–8).

This finding supports the notion that the hydrogenation of CO_2 towards *CO formation favours ER-type coupling with solvent protons, in contrast to the hydrogenation of SO_2/CO_2 towards *S and formate production that follows LH-type coupling with adsorbed H (Fig. 1d). We also found that the presence of SO_2 greatly decreases the *COOH surface coverage (Supplementary Fig. 9 and Supplementary Tables 3 and 4). The *SOOH and *HSO_2 intermediates produced by SO_2 reduction showed stronger binding on Cu compared to other hydrogenated intermediates, indicating that these are preferred reducing species under reductive potential (Supplementary Fig. 10) and that the hydrogenation of SO_2 via adsorbed H is favoured over CO_2 reduction, leading to a build-up of adsorbed H on the Cu surface.

We then varied the *H coverage ranging from 0 to 4/9 monolayer (ML) (fraction of active sites occupied by the species of interest) and calculated the free energy changes of *COOH, *OCHO and *H, and the related surface free energy changes on Cu as a function of applied potential (Supplementary Fig. 11). We found that increasing *H coverage inhibits *COOH to depromote its reaction; however, it weakens *OCHO and *H to facilitate their ongoing reduction. This is because Cu weakly adsorbs COOH while strongly interacting with *OCHO and *H. To validate, we further calculated the free energy changes of the key elementary steps (ΔG_{key}) concerning (1) *COOH formation towards *CO production and (2) the desorption and production of HCOOH and H_2, respectively, under various *H coverages (Fig. 2d and Supplementary Figs. 12–14). An increase in ΔG_{key} indicates that the catalytic reaction is energetically less favoured. The DFT results show that the increase in *H coverage can inhibit CO_2R towards *CO while promoting HER and CO_2R towards HCOOH. We concluded that increasing the ratio of solvent H to adsorbed H on Cu would simultaneously deactivate SO_2 hydrogenation and suppress H_2 and formate production, thereby facilitating *CO formation and C_{2+} production.

https://doi.org/10.1038/s41560-024-01577-9

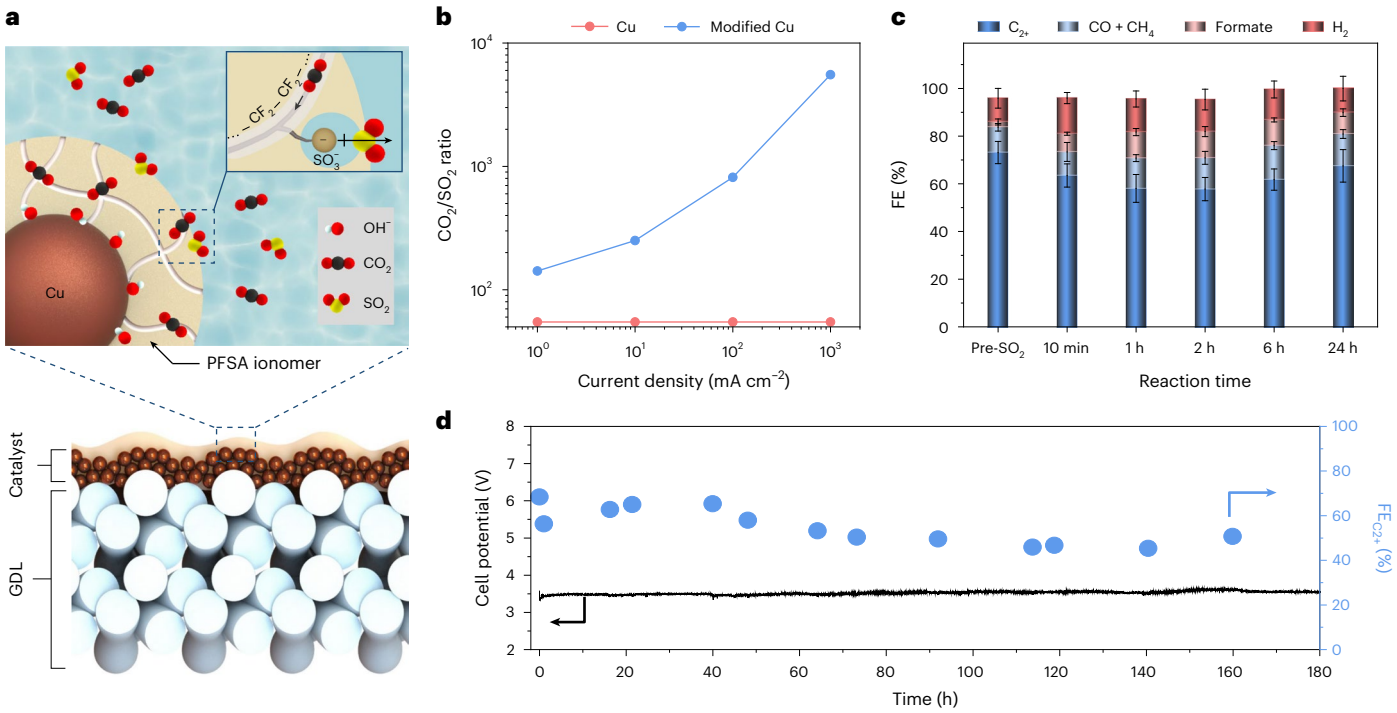

Fig. 3 | SO_2-tolerant CO_2 electrolysis at a modified planar Cu electrode. **a**, Schematic of a planar PFSA@Cu/PTFE heterojunction with hydrophilic and hydrophobic characteristics provided by $-SO_3^-$ and $-CF_2$ functionalities for interacting with SO_2 and facilitating CO_2 transport, respectively. **b**, Modelled mass flux ratios of CO_2 over SO_2 on the ionomer-modified Cu and bare Cu electrodes at different reaction rates during the co-electrolysis of CO_2 and SO_2. **c**,**d**, Product analysis (**c**) and extended operational stability (**d**) of the PFSA@Cu/PTFE electrode. All experiments were performed at a constant current density of 100 mA cm^{-2} using a gas feed comprising dilute CO_2 (50%, balanced with N_2) and 400 ppm SO_2. Where error bars are shown, values are means, and error bars indicate s.d. ($n = 3$ replicates).

Impact of hydrophobicity on SO_2 poisoning

Motivated by the DFT results, we sought to design experimentally a Cu catalyst with controlled surface water availability. We turned our attention to tuning the polytetrafluoroethylene (PTFE) loading in the gas diffusion electrode, as it offers control of the water and gas transport over the gas diffusion layer (GDL)[39,40]. We tested sputtered Cu catalysts supported by GDLs with PTFE loadings ranging from 25 wt% to 100 wt% at 100 mA cm^{-2} over 6 h using a dilute CO_2 gas with 400 ppm SO_2. We found that the surface wettability of the same Cu electrode gradually increased with the reaction time and that increasing the PTFE loading in the Cu electrode helps preserve surface hydrophobicity during SO_2/CO_2 co-electrolysis (Fig. 2e). Concurrently, we observed a decrease in H_2 and formate FE values and enhanced FE_{C2+} with increased PTFE loading (Fig. 2f and Supplementary Fig. 15). The Cu catalyst deposited onto a pure PTFE GDL showed a FE_{C2+} of >40% over 6 h of SO_2/CO_2 co-electrolysis. However, after over 24 h of electrolysis, this selectivity drops to 30% as the hydrogen FE (FE_{H2}) increases to 40% (Supplementary Fig. 16), suggesting that the SO_2 poisoning was mitigated but not fully suppressed.

To gain insight into the SO_2 poisoning mechanism, we characterized the electrodes with different PTFE loadings before and after adding SO_2 in the CO_2 feed using in situ Raman spectroscopy. The 90% PTFE-supported Cu electrode showed that before the addition of SO_2, the Cu catalysts exhibited strong characteristic signals at 280 cm^{-1} (characteristic of the frustrated rotational mode of CO) and 360 cm^{-1} (associated with Cu–CO stretching; Supplementary Fig. 17). However, when SO_2 was added, these two peaks disappeared over a 10 min SO_2 exposure, indicating that SO_2 reduction at the Cu surface eliminates the *CO coverage. As a result, *H coverage is increased, consistent with an immediate drop in FE_{C2+} and a corresponding increase in FE_{H2} over the 10 min SO_2 exposure time (Supplementary Fig. 17c). By contrast, under the same testing conditions, the *CO peak broadened, with its intensity gradually decreasing on the pure PTFE-supported Cu electrode with the addition of SO_2 (Supplementary Fig. 18). This finding suggests that the catalyst is poisoned by SO_2, likely with its reduction products accumulated on the Cu surface, diminishing *CO adsorption and the FE_{C2+} from CO_2R.

To probe the SO_2 reduction process on Cu, we removed SO_2 from the gas feed but instead introduced 400 ppm SO_3^{2-} (the dissolved form of SO_2 in the electrolyte) or S^{2-} (the most reduced product from SO_2 reduction) in the electrolyte. No poisoning was observed, and CO_2R performance was unchanged in both cases (Supplementary Fig. 19). Ex situ characterizations after SO_2/CO_2 co-electrolysis, including transmission electron microscopy (TEM) and X-ray photoemission spectroscopy, evidenced the presence of Cu-based oxide, sulfide and sulfate in addition to elemental S and metallic Cu species (Supplementary Figs. 20 and 21). However, no high-valence Cu species were detected from the in situ Raman spectroscopy (Supplementary Fig. 22) and in situ X-ray absorption spectroscopy (XAS; Supplementary Figs. 23 and 24) measurements. The results suggest that SO_2 poisoning does not modulate the oxidation state or coordination environment of Cu.

Taken together, these results show that the Cu remains metallic during SO_2/CO_2 co-electrolysis, and *S can reside on the Cu surface to impact the long-term CO_2 electrolysis. Additional DFT calculations of the ΔG_{key} relating to the *COOH formation and the desorption of HCOOH and H_2 support the view that increasing *S coverage inhibits *COOH adsorption and thus *CO production, but weakens *H and *OCHO adsorption to facilitate H_2 and formate production on Cu under applied negative potentials (Supplementary Note 3 and Supplementary Figs. 25–28).

Investigation of the planar polymer/catalyst/ionomer catalyst

We sought to further increase C_{2+} selectivity by limiting the SO_2 mass transport, as a means to circumvent the accumulation of *S on Cu

https://doi.org/10.1038/s41560-024-01577-9

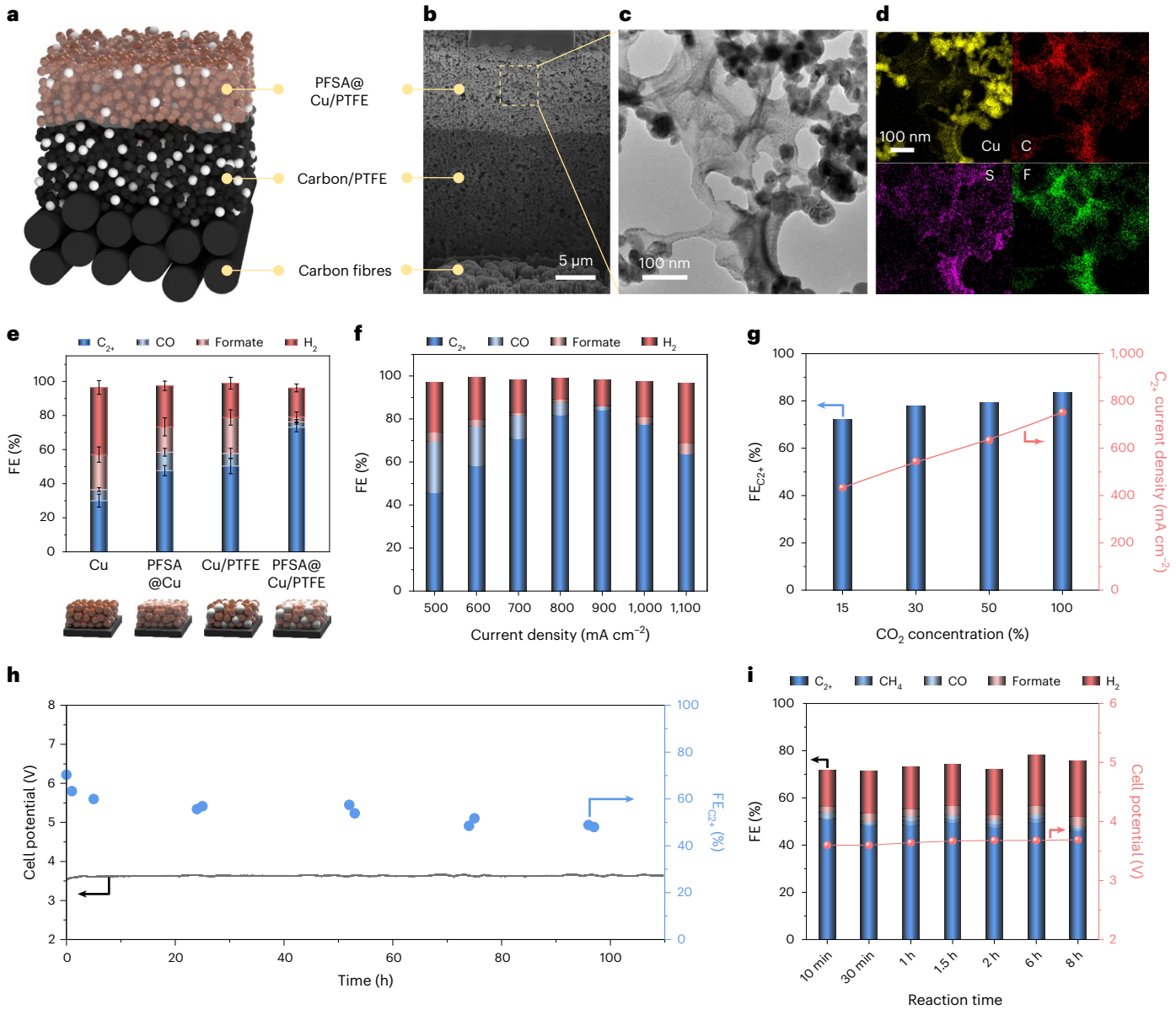

Fig. 4 | Structure and performance of a modified bulk Cu electrode for the co-electrolysis of CO_2 and SO_2. **a**, Schematic of the bulk PFSA@Cu/PTFE electrode. **b**, Cross-sectional scanning electron microscopy image of a PFSA@Cu/PTFE electrode fabricated with a focused-ion beam. **c**,**d**, TEM image of a PFSA@Cu/PTFE laminar fabricated with a focused-ion beam (**c**) and the relevant elemental mapping of Cu (yellow), C (red), S (purple) and F (green; **d**). **e**, Product selectivities from the co-electrolysis of dilute CO_2 (50%, balanced with N_2) and 400 ppm SO_2 on bare Cu; PFSA-modified Cu (PFSA@Cu); PTFE-modified Cu (Cu/PTFE); and PFSA@Cu/PTFE at 400 mA cm^{-2}. Values are means, and error bars indicate s.d. ($n = 3$ replicates). Insets at the bottom show schematics of the various species. **f**, Product selectivities from the co-electrolysis of 100% CO_2 and 400 ppm SO_2 on PFSA@Cu/PTFE at various current densities. **g**, Effect of CO_2 concentration on C_{2+} product selectivities and partial current densities from the co-electrolysis of CO_2 (15–100%, balanced with N_2) and 400 ppm SO_2 on PFSA@Cu/PTFE. **h**, Performance of the PFSA@Cu/PTFE electrode at 250 mA cm^{-2} using a dilute CO_2 source (50%, balanced with N_2) with 400 ppm SO_2. **i**, Performance of the PFSA@Cu/PTFE electrode at 200 mA cm^{-2} using a simulated flue gas feed comprising 15% CO_2, 4% O_2, 500 ppm NO and 500 ppm SO_2 (balanced with N_2).

from SO_2 reduction. We sought a material strategy that would diminish SO_2 mass transport to the Cu surface while enabling unimpeded CO_2 transport. For this, we employed an adlayer of perfluorosulfonic acid (PFSA) ionomer with combined hydrophobic and highly charged hydrophilic domains (Fig. 3a)[25,41,42]. The hydrophilic domain of PFSA interacts strongly with polar SO_2 and inhibits its mass transport through the PFSA, resulting in a high SO_2 solubility (~30 times higher than the solubility of non-polar CO_2) and a low SO_2 diffusivity (~5 times lower than CO_2 diffusivity) in the PFSA[36,41,43]. Conversely, the hydrophobic tetrafluoroethylene chains provide an unimpeded route for non-polar CO_2, boosting its mass transport[25]. In addition, the hydrophobic backbone together with the negatively charged side chains attached to the Cu surface creates a confinement layer to retain the locally generated hydroxide ions via Donnan exclusion. The combined effect is to diminish *H surface coverage, disfavour SO_2 activation and facilitate C–C coupling for CO_2 reduction[41].

To assess the impact of the ionomer on CO_2 and SO_2 gas availability, we carried out a reaction–diffusion model analysis[44,45], in which we introduced a PFSA adlayer at the catalyst surface (Supplementary Fig. 29; details in Supplementary Note 4). Different solubilities and gas diffusion coefficients of CO_2 and SO_2 in PFSA were implemented (Supplementary Tables 5 and 6). Modelling the mass transport of CO_2 and trace SO_2 on different Cu electrodes indicates that the adlayer generally promotes CO_2/SO_2 mass flux ratios on the Cu surface (Fig. 3b and

Supplementary Fig. 30), owing to the decreased SO_2 transport through the hydrophilic domains compared to the improved transport of CO_2 through the hydrophobic domains. As the operating current density increases (from 1 mA cm^{-2} to 1 A cm^{-2}), hydroxide ions generated at the cathode are concentrated due to Donnan exclusion, which limits SO_2 mass transport due to the Sechenov law (Supplementary Fig. 31)[41]. As a result the Cu catalyst with the adlayer exhibited a two orders of magnitude higher CO_2/SO_2 mass flux ratio compared with bare Cu.

We applied a thin adlayer of PFSA ionomer at a thickness of ~10 nm onto the Cu/PTFE electrode (Supplementary Fig. 32), forming a planar PFSA@Cu/PTFE electrode. At a constant operating current density of 100 mA cm^{-2}, the PFSA@Cu/PTFE electrode showed a slight increase in the formate FE ($FE_{formate}$) from 2% to ~10% within the first 10 min of exposure to 400 ppm SO_2. It then exhibited a sustained 60% FE_{C2+} for 24 h (Fig. 3c). In situ Raman measurements showed that both the peak intensity and vibrational frequencies of those characteristic *CO peaks on Cu were unchanged after adding SO_2 (Supplementary Fig. 33). The improved CO_2R performance is attributed to a tenfold increase in the CO_2/SO_2 mass flux ratio (Fig. 3b) that diminishes SO_2 poisoning and locally confined hydroxide ions that favour C_{2+} production. This approach achieved stable CO_2R in the presence of 400 ppm SO_2 (Fig. 3d): the FE towards C_{2+} surpassed 50% for an initial 160 h at a full-cell potential of −3.5 V (without *iR* compensation; *i*, current; *R*, resistance).

The effect of the PFSA on CO_2R activity was not substantial; the PFSA@Cu/PTFE electrode showed product distribution akin to bare Cu before adding SO_2, and the addition of the PFSA adlayer did not change the surface wettability or the electrochemical surface area of the Cu appreciably (Supplementary Fig. 34). It is also unlikely that PFSA affects the electronic structure of Cu: only metallic Cu was observed by operando XAS (Supplementary Fig. 24).

The addition of PFSA is expected to influence the local transport of gas reactants, water and ion species in electrolysis[25,41,46]. To experimentally explore the relationship between the adlayer properties and SO_2 tolerance, we varied the density of hydrophilic sulfonic groups in PFSA, thereby tuning CO_2/H_2O transport to the Cu surface (Supplementary Fig. 35)[41]. We found that increasing H_2O transport facilitated HER and lowered SO_2 tolerance. To assess the role of the PFSA hydrophobic domains, we applied an adlayer of perfluorocarbon with a solely hydrophobic backbone akin to PFSA as the control (Supplementary Fig. 36). This adlayer-modified electrode exhibited a low HER ($FE_{H2} \approx 15\%$) similar to PFSA@Cu/PTFE, but a high formate production ($FE_{formate} \approx 30\%$) over 2 h of SO_2/CO_2 co-electrolysis. This response is attributed to the enhancement of CO_2 and SO_2 transport and lowered H_2O transport, resulting in SO_2 poisoning that promotes formate generation from CO_2R.

To examine the role of the PFSA amphiphilic property, we added dodecyltrimethylammonium bromide in the electrolyte[21]. This amphiphilic molecule reduced FE_{C2+} by promoting CO formation before adding SO_2 and resulted in SO_2 poisoning with enhanced productions of H_2 and formate (Supplementary Fig. 37). The PFSA adlayer could also contribute to the enrichment of potassium ions (K^+) via its sulfonic groups to accelerate CO_2 activation[46]. To assess the impact of K^+ concentration on SO_2 tolerance, we tuned the concentrations of K^+ ions in the electrolyte (Supplementary Fig. 38) and found that enriching K^+ on Cu resulted in increased HER and SO_2 poisoning. These results further confirm that it is the enhanced CO_2/SO_2 flux ratio property of the ionomer that mitigates SO_2 poisoning.

These findings suggest a SO_2 tolerance strategy wherein the Cu catalyst is contained between the PTFE layer and the PFSA ionomer adlayer to diminish *H coverage and SO_2 transport, and thereby deactivate SO_2 hydrogenation and inhibit the LH-type CO_2 hydrogenation towards formate. Concurrently, the strategy enables abundant CO_2 diffusion towards the interlayer catalyst and prioritizes the ER-type CO_2 hydrogenation towards *CO formation and C_{2+} production (Fig. 3a). The low thickness of the SO_2 rejection layer suggests that the strategy is not limited to the planar catalytic surfaces, and rather could be extended to a bulk heterojunction that could offer increased reaction rates and further improved performance (Fig. 3b).

Investigation of the bulk polymer/catalyst/ionomer catalyst

We constructed a bulk PFSA@Cu/PTFE electrode comprising PFSA ionomers, porous Cu nanoparticles and PTFE nanoparticles to increase the electrochemical surface area of the electrode while enabling SO_2 tolerance throughout (Fig. 4a–d and Supplementary Figs. 34 and 39). Similar to the case of planar PFSA@Cu/PTFE, the SO_2 tolerance of bulk PFSA@Cu/PTFE was dependent on the combination of PTFE and PFSA (Fig. 4e). The optimized bulk PFSA@Cu/PTFE electrode showed a peak FE_{C2+} of 84% over a wide range of operating current densities from 500 mA cm^{-2} to 1,100 mA cm^{-2} (Fig. 4f), with CO being the other CO_2R product. Based on the model presented herein, the observed enhancement can be explained by the ~100 times increased diffusion of CO_2 relative to SO_2 (Fig. 3b).

To challenge the versatility of the bulk PFSA@Cu/PTFE, we assessed CO_2R performance using gas streams with 400 ppm SO_2 and different CO_2 concentrations ranging from 15% to 100%, a spectrum that covers both raw and scrubbed flue gas streams (Fig. 4g)[29,30,47,48]. We found that the PFSA@Cu/PTFE electrodes were SO_2 tolerant at all CO_2 concentrations. Reducing the CO_2 concentration induced a slight decrease in current density, a finding attributable to reduced CO_2 availability at the catalyst layer.

Tuning conditions for high FE_{C2+} and low cell potential, we achieved a C_{2+} energy efficiency (EE_{C2+}) of ~23% (without *iR* compensation) over a range of current densities (600 mA cm^{-2} to 1,000 mA cm^{-2}) and over a broad range of input CO_2 concentrations (15% to 100%; Supplementary Table 7). We achieved a high FE_{C2+} of up to 84% (exceeding 70% throughout the range of conditions tested), at a current density of 900 mA cm^{-2}, by modulating the CO_2 concentration (via the input flow-rate) and the applied potential or current density. A high FE_{C2+} of 79% was achieved at 1,000 mA cm^{-2}; this combination of current density and selectivity results in a high C_{2+} productivity of 790 mA cm^{-2}. This represents an efficient SO_2-tolerant CO_2R system towards C_{2+} production, in contrast to the prior impurity-added CO_2R systems (Supplementary Table 8)[8,21], and outperforms literature benchmark CO_2R systems achieved with pure CO_2 input (Supplementary Table 9). Additional technoeconomic analysis for different CO_2R systems (Supplementary Note 6, Supplementary Figs. 48 and 49 and Supplementary Table 10) highlights the prospects of achieving economically viable C_2 production (for example, ethanol) via flue gas electrolysis.

Using a dilute CO_2 source (50%, balanced with N_2) with the presence of 400 ppm SO_2, we operated the planar PFSA@Cu/PTFE electrode at a current density of 250 mA cm^{-2} stably for 100 h, and the FE towards C_{2+} was above 50% throughout (Fig. 4h). Even beyond 100 h at high current, SO_2 poisoning was not an observed failure mode, and rather failure was caused by the accumulation of salt in the gas channel or related flooding of the electrode, an established challenge for CO_2 electrolysers (Supplementary Fig. 40).

To decouple the influences of flooding and poisoning, we modulated the humidity of gas feedstocks. We found that an increase of humidity in the gas stream, without SO_2, led to an increase of FE_{H2} due to flooding, whereas the system showed a substantial increase in both FE_{H2} and $FE_{formate}$ with SO_2 addition due to the combination of poisoning and flooding (Supplementary Figs. 41 and 42). As an additional control, we tested at a high electrolyte concentration, as susceptibility to flooding is known to increase with electrolyte concentration. Similar to the humidity tests, SO_2 poisoning was indicated by increased $FE_{formate}$, and the rate of poisoning was accelerated by conditions leading to increased wetting of the electrode (Supplementary Figs. 43 and 44). We concluded that electrode flooding could lead to increased hydrogen

coverage on the catalyst, accelerating SO_2 activation and poisoning, and therefore a consistently dry environment is required to prevent SO_2 poisoning.

Investigation of the simulated industrial flue gases

In addition to SO_2, it is common for major industrial point sources to also contain NO, O_2, N_2 and lower concentrations of CO_2. The ability to react this challenging combination would eliminate post-capture steps and enable direct electrocatalytic production of multi-carbon products from the most prevalent and challenging CO_2 point sources. We tested the PFSA@Cu/PTFE strategy with a simulated flue gas comprising 15% CO_2 in the presence of common contaminants such as NO, O_2 and SO_2 (refs. 8,21,49,50).

Adding 500 ppm NO in the 15% CO_2 stream did not result in an observable change in CO_2R activity due to the limited transport of NO, and we saw a lack of NO poisoning on Cu (Supplementary Fig. 45), consistent with a previous report[50]. By introducing 4% O_2 in the dilute CO_2 stream[51], we achieved, at a current density of 200 mA cm^{-2}, an FE of 55% for C_{2+} products on planar PFSA@Cu/PTFE (Supplementary Fig. 46). By contrast, a low FE_{C2+} of 20% was observed on bare Cu, in line with our DFT results, indicating that the reduction of O_2 (favoured over CO_2R) and the adsorption of O_2 and its reduced species (for example, *OH) hinder C–C coupling on Cu (Supplementary Note 5 and Supplementary Fig. 47). The addition of this adlayer, therefore, not only promotes the CO_2/SO_2 mass flux ratio, but boosts CO_2 over O_2 transport to enhance C_{2+} production. Feeding a simulated flue gas comprising 15% CO_2, 4% O_2, 500 ppm NO and 500 ppm SO_2, this approach achieves a sustained FE_{C2+} of ~50% at 200 mA cm^{-2} (Fig. 4i).

Conclusions

This work demonstrates that CO_2 electrolysis can not only proceed in the presence of catalyst-poisoning impurities typical of point sources, but it can do so with performance and stability levels that are among the best in the field. We investigated the mechanisms governing SO_2 contamination of CO_2R catalysts and applied the results to develop a catalyst resistant to poisoning. We extended this approach to high-surface-area catalysts to demonstrate impurity resistance while operating at high energy efficiency and current density. The direct conversion of flue gas, at high current density, addresses both capture and capital costs and thereby removes two key barriers to industrial CO_2 conversion.

Methods

Materials and chemicals

Chemical precursors used for catalyst, electrolyte and electrode synthesis/preparation were purchased from Sigma-Aldrich. These include high-purity potassium bicarbonate, PFSA-1000 (Nafion, D520/D521), PFSA-790 (Aquivion, D79), copper nitrate trihydrate, copper(II) oxide powder, copper(I) sulfide powder, copper(II) sulfate powder, sodium borohydride, 60% PTFE dispersion in H_2O, Triton X-100, isopropanol and carbon nanoparticles. An X37-50 Sustainion anion exchange membrane was purchased from Dioxide Materials. The iridium oxide anode was purchased from Magneto Special Anodes. The PTFE GDL with 450 nm pore size was purchased from Beijing Zhongxingweiye Instrument Co. The carbon paper GDLs used were the MGL 190 with 50% wet proofing electrodes purchased from Fuel Cell Earth.

Sample fabrication

Fabrication of catalysts with the carbon paper or PTFE GDLs. Carbon paper GDLs with varying PTFE percent content in their microporous layer were fabricated by blade coating a 0.5 mm layer of mixed PTFE and carbon nanoparticle solutions over a carbon paper GDL. The carbon nanoparticle solution was prepared by mixing 0.4 g of carbon nanoparticles with 35 ml of isopropanol and four drops of Triton X-100. The PTFE solution was prepared by mixing X g of 60% PTFE dispersion with Y g of deionized water. The X g of PTFE dispersion was chosen based on the desired final PTFE content in the substrate, with $X + Y = 10$ g. The total electrode area for blade coating was 15 × 20 cm^2. After blade coating, the substrates were air-dried for 3 h before being placed in an oven for 3 h at 280 °C and 1 h at 340 °C. The lowest PTFE content in the microporous layer was 25%, which is considered the base case for the carbon paper GDL. Copper was then sputtered onto the carbon paper and PTFE substrates using an Angstrom Nexdep sputtering system. A 99.99% pure Cu target was used in a vacuum environment (~10^{-6} torr) at a deposition rate of 1 Å s^{-1}. The thickness of the sputtered copper was kept constant at 200 nm for all experiments.

PFSA@Cu/PTFE fabrication. The planar PFSA@Cu/PTFE was fabricated by spray-coating 3 µl of 5 wt% Nafion solution dispersed in 2 ml of methanol onto a 3 cm × 1.5 cm sample of the Cu catalyst with the PTFE GDL. The catalyst was subsequently vacuum dried for 3 h prior to the experiments. Control electrodes included spray-coating loadings ranging between 0.5–3 µl cm^{-2} of 5 wt% PFSA-1000-based PFSA@Cu/PTFEs and 0.1–0.4 µl cm^{-2} of 25 wt% PFSA-790 onto the Cu catalyst with the PTFE GDL. For the fabrication of the bulk PFSA@Cu/PTFE electrode, porous copper nanoparticles were synthesized through the aqueous reduction of copper(II) nitrate trihydrate by sodium borohydride. First, 1,820 mg of copper nitrate trihydrate powder was added to 400 ml of deionized water. The clear solution was then transferred to an ice bath and stirred for 15 min. Next, 7,700 mg of sodium borohydride was added to 40 ml of ice-cooled deionized water and shaken until clear. The sodium borohydride solution was then added to the ice-cooled copper nitrate trihydrate solution and stirred for 5 min until discernible black flocs became visible. The solution was then removed from the ice bath and placed in a dry environment for 5 h. Lastly the solutions were decanted with ethanol and dried in a centrifuge four times to collect the porous copper nanoparticles. The PFSA@Cu/PTFE electrodes were then fabricated by spray-coating a copper-based dispersion on the 39 BB Sigracet carbon paper GDL to a loading of 1 mg cm^{-2}. The dispersion consisted of 50 mg copper nanoparticles, 30 mg dispersed PTFE solution and 11 µl of 5 wt% Nafion solution dispersed in 4 ml isopropanol. The catalyst was subsequently vacuum dried for 3 h prior to the experiments.

Membrane electrode assembly set-up

All experiments were performed using a neutral membrane electrode assembly electrolyser. The cell was assembled using the same set-up as in ref. 35 but with an active area of 1 cm^2 for each catalyst. The 0.1 M $KHCO_3$ electrolyte was pumped at a flow-rate of 0.5 ml min^{-1} via a peristaltic pump.

The SO_2-free experiments were operated by mixing pure CO_2 and N_2 gas at flow-rates of 14 and 16 sccm, respectively, to produce a final flow-rate of 30 sccm with 46% CO_2. The SO_2 experiments mixed a gas stream with a composition of 600 ppm SO_2, 19.94% CO_2 and 80% N_2 at 20 sccm, with a stream of pure CO_2 flowing at 10 sccm. This produced a final flow-rate of 30 sccm, with 46% CO_2 and 400 ppm SO_2. To achieve other final gas compositions, the pure CO_2, pure N_2 and mixed $SO_2/CO_2/N_2$ gas lines were always combined in a manner that produced a final flow-rate of 30 sccm and 46% CO_2 inlet stream. The gases were flowed using a Sierra SmartTrak 100 mass flow controller, and the gas cylinders were purchased from Praxair.

Electrochemical measurements

Electrochemical experiments were performed using an electrochemical workstation (Autolab PGSTAT204) with a current booster (10 A FRA32). All reported cell voltages were not *iR* corrected. The experiments were performed at a constant current density of 100 mA cm^{-2} unless otherwise noted. The system was always run for 30 min before adding SO_2. The purpose of this was to ensure that the system has already reached a steady state once SO_2 was added.

Product analysis

Gas samples were collected at the outlet of the cathode gas channel using a gas syringe. Some 1 ml of gas sample was injected into a gas chromatograph (PerkinElmer Clarus 590). The gas chromatograph consisted of a flame ionization detector and a thermal conductivity detector. The gas chromatograph used a Molecular Sieve 5A capillary column and a packed Carboxen-1000 column from Sigma-Aldrich with Ar carrier gas. The FE of H_2, CO, CH_4 and C_2H_4 could be calculated using the following equation:

$$\mathrm{FE}\,(\%) = x_\mathrm{i} \times v \times \frac{n_\mathrm{i} F P_\mathrm{o}}{R T I_\mathrm{total}} \times 100$$

where x_i is the volume fraction of gas product i, v is the gas flow-rate at the cathode gas outlet, n_i is the number of electrons transferred to produce one molecule of product i, F is the Faraday constant, P_o is the atmospheric pressure, T is the temperature, R is the ideal gas constant and I_total is the total current.

Liquid samples were collected through a cold trap at the cathode gas outlet, and were combined with liquid samples from the electrolyte in order to account for any liquid product crossover. The concentration of liquid products was measured using ^{1}H NMR spectroscopy on an Agilent DD2 500 MHz spectrometer. The reference standard was dimethyl sulfoxide, with D_2O used as water suppression. The liquid products measured were formate, acetate, ethanol and propanol.

Characterization

Materials characterization. Scanning electron microscopy imaging was performed using a high-resolution scanning electron microscope (Hitachi S-5200). TEM measurements and associated energy dispersive X-ray elemental mapping were carried out via a Hitachi HF-3300 instrument. The cross-sectional lamella was prepared by focused-ion beam milling and thinning processes using a gallium ion beam and attached to a TEM stage for scanning TEM and energy dispersive X-ray analysis. X-ray photoemission spectroscopy measurements for surface element investigation were carried out using a PHI 5700 Electron Spectroscopy for Chemical Analysis system.

In situ Raman spectroscopy measurements. In situ Raman spectroscopy was performed using an electrolyte reservoir electrolyser to study the catalysts and substrates under SO_2 conditions. The 200 nm sputtered copper catalysts over the varying gas diffusion substrates were used to separate the electrolyte from the gas channel. Some 400 ppm SO_2 and 50% CO_2 balanced with N_2 were flowed in the gas compartment. A Ag/AgCl reference electrode and Pt wire counter electrode were placed in the 0.1 M $KHCO_3$ electrolyte. The cathode area was 1 cm^2. A ×63 immersion objective lens was used to collect Raman spectra with 10 s exposure time. The configuration used a Renishaw Raman spectrometer with a 785 nm laser, 1,200 mm^{-1} grating and 200 μW laser power.

In situ X-ray absorption spectroscopy measurements. In situ XAS measurements of the Cu K edge were performed at the Soft X-ray Microcharacterization Beamline of the Canadian Light Source. The XAS spectra were recorded in a fluorescence yield mode with a standard metallic Cu foil applied for energy calibration (to 8,979 eV). During measurements, 0.1 M $KHCO_3$ aqueous solution was applied for both catholyte and anolyte and an anion exchange membrane (Fumasep FAA-3-PK-130) was used to separate the catholyte and anolyte compartments. A sputtered Cu (200 nm) catalyst loaded on carbon paper, a sputtered Pt (200 nm) catalyst on a microporous PTFE gas diffusion substrate and a Ag/AgCl wire (saturated KCl) were used as working, counter and reference electrodes, respectively. The electrolytes were circulated through the in situ XAS cell using peristaltic pumps at a flow-rate of 0.5 ml min^{-1}. A simulated flue gas consisting of 400 ppm SO_2 and 46% CO_2 (balanced with N_2) was supplied in the gas compartment during the test. An Autolab PGSTAT204 in galvanostatic mode was used as an electrochemical workstation. XAS data were analysed using the Athena and Artemis software programs included in a standard IFEFFIT package.

Contact angle measurements. The apparent contact angles of the GDLs were measured before and after the electrolysis experiments to study the changes in wettability of the CO_2R catalyst due to the presence of SO_2. The GDL samples measured after the electrolysis experiments were first washed with deionized water and dried with nitrogen gas. The samples were then placed in a sealed vacuum chamber for 24 h prior to the measurements. The apparent contact angles were measured using a Ramé-hart contact angle goniometer (model 181060). The samples were taped on the stage of the contact angle goniometer to avoid observing curvature on the contact line. Next, 10–15 μl deionized water droplets were deposited onto the catalyst surface using a Gilmont micro syringe, and the apparent contact angles were measured.

Density functional theory calculations

All first-principles DFT calculations were conducted using the Vienna Ab initio Simulation Package[52,53]. The generalized gradient approximation in the Perdew–Burke–Ernzerhof functional[54] was used for the electron exchange and correlation interaction, while the projector augmented-wave method[55] was chosen to describe the ion–electron interactions. The calculations were performed using a plane-wave basis set within a kinetic energy cut-off of 450 eV. The DFT-D3 method developed by Grimme et al. was employed to describe the long-range dispersion interaction[56]. The k-point sampling of a 3 × 3 × 1 mesh within the Monkhorst–Pack scheme was used for the optimization[57]. For the metal system, the second-order Methfessel–Paxton smearing σ of 0.2 eV was employed with a width of 0.2 eV to determine partial electron occupancies[58]. The calculations for optimization were completed when the force and energy of the self-consistent electronic iteration reach 0.02 eV Å^{-1} and 10^{-5} eV, respectively. Calculation details are included in Supplementary Note 2.

Data availability

The datasets analysed and generated during the current study are included in the paper and its Supplementary Information. Source data are provided with this paper.

References

1. Jordaan, S. M. & Wang, C. Electrocatalytic conversion of carbon dioxide for the Paris goals. *Nat. Catal.* **4**, 915–920 (2021).
2. Sullivan, I. et al. Coupling electrochemical CO_2 conversion with CO_2 capture. *Nat. Catal.* **4**, 952–958 (2021).
3. Nitopi, S. et al. Progress and perspectives of electrochemical CO_2 reduction on copper in aqueous electrolyte. *Chem. Rev.* **119**, 7610–7672 (2019).
4. Ozden, A. et al. Carbon-efficient carbon dioxide electrolysers. *Nat. Sustain.* **5**, 563–573 (2022).
5. Hepburn, C. et al. The technological and economic prospects for CO_2 utilization and removal. *Nature* **575**, 87–97 (2019).
6. Li, M., Irtem, E., van Montfort, H. P. I., Abdinejad, M. & Burdyny, T. Energy comparison of sequential and integrated CO_2 capture and electrochemical conversion. *Nat. Commun.* **13**, 5398 (2022).
7. Welch, A. J., Dunn, E., DuChene, J. S. & Atwater, H. A. Bicarbonate or carbonate processes for coupling carbon dioxide capture and electrochemical conversion. *ACS Energy Lett.* **5**, 940–945 (2020).
8. Luc, W. et al. SO_2-induced selectivity change in CO_2 electroreduction. *J. Am. Chem. Soc.* **141**, 9902–9909 (2019).
9. De Luna, P. et al. What would it take for renewably powered electrosynthesis to displace petrochemical processes? *Science* **364**, eaav3506 (2019).
10. Keith, D. W., Holmes, G., Angelo, D. S. & Heidel, K. A process for capturing CO_2 from the atmosphere. *Joule* **2**, 1573–1594 (2018).

11. Sanz-Pérez, E. S., Murdock, C. R., Didas, S. A. & Jones, C. W. Direct capture of CO_2 from ambient air. *Chem. Rev.* **116**, 11840–11876 (2016).
12. Sharifian, R., Wagterveld, R., Digdaya, I., Xiang, C. & Vermaas, D. Electrochemical carbon dioxide capture to close the carbon cycle. *Energy Environ. Sci.* **14**, 781–814 (2021).
13. Li, Y. C. et al. CO_2 electroreduction from carbonate electrolyte. *ACS Energy Lett.* **4**, 1427–1431 (2019).
14. Li, T. et al. Electrolytic conversion of bicarbonate into CO in a flow cell. *Joule* **3**, 1487–1497 (2019).
15. Li, T., Lees, E. W., Zhang, Z. & Berlinguette, C. P. Conversion of bicarbonate to formate in an electrochemical flow reactor. *ACS Energy Lett.* **5**, 2624–2630 (2020).
16. Lees, E. W. et al. Electrolytic methane production from reactive carbon solutions. *ACS Energy Lett.* **7**, 1712–1718 (2022).
17. Lees, E. W. et al. Electrodes designed for converting bicarbonate into CO. *ACS Energy Lett.* **5**, 2165–2173 (2020).
18. Zhang, Z. et al. Porous metal electrodes enable efficient electrolysis of carbon capture solutions. *Energy Environ. Sci.* **15**, 705–713 (2022).
19. Lee, G. et al. Electrochemical upgrade of CO_2 from amine capture solution. *Nat. Energy* **6**, 46–53 (2021).
20. Zhang, Z. et al. Conversion of reactive carbon solutions into CO at low voltage and high carbon efficiency. *ACS Cent. Sci.* **8**, 749–755 (2022).
21. Pimlott, D. J. D., Jewlal, A., Mowbray, B. A. W. & Berlinguette, C. P. Impurity-resistant CO_2 reduction using reactive carbon solutions. *ACS Energy Lett.* **8**, 1779–1784 (2023).
22. Lee, G. et al. CO_2 electroreduction to multicarbon products from carbonate capture liquid. *Joule* **7**, 1277–1288 (2023).
23. Shin, H., Hansen, K. U. & Jiao, F. Techno-economic assessment of low-temperature carbon dioxide electrolysis. *Nat. Sustain.* **4**, 911–919 (2021).
24. Realmonte, G. et al. An inter-model assessment of the role of direct air capture in deep mitigation pathways. *Nat. Commun.* **10**, 3277 (2019).
25. de Arquer, F. P. G. et al. CO_2 electrolysis to multicarbon products at activities greater than 1 A cm^{-2}. *Science* **367**, 661–666 (2020).
26. Wen, G. et al. Continuous CO_2 electrolysis using a CO_2 exsolution-induced flow cell. *Nat. Energy* **7**, 978–988 (2022).
27. Li, W. et al. Bifunctional ionomers for efficient co-electrolysis of CO_2 and pure water towards ethylene production at industrial-scale current densities. *Nat. Energy* **7**, 835–843 (2022).
28. Verma, S., Lu, S. & Kenis, P. J. Co-electrolysis of CO_2 and glycerol as a pathway to carbon chemicals with improved technoeconomics due to low electricity consumption. *Nat. Energy* **4**, 466–474 (2019).
29. Wang, X. & Song, C. Carbon capture from flue gas and the atmosphere: a perspective. *Front. Energy Res.* **8**, 560849 (2020).
30. Husebye, J., Brunsvold, A. L., Roussanaly, S. & Zhang, X. Techno economic evaluation of amine based CO_2 capture: impact of CO_2 concentration and steam supply. *Energy Procedia* **23**, 381–390 (2012).
31. Wei, P. et al. Coverage-driven selectivity switch from ethylene to acetate in high-rate CO_2/CO electrolysis. *Nat. Nanotechnol.* **18**, 299–306 (2023).
32. Gao, J., Wang, S., Zhao, B., Qi, G. & Chen, C. Pilot-scale experimental study on the CO_2 capture process with existing of SO_2: degradation, reaction rate, and mass transfer. *Energy Fuels* **25**, 5802–5809 (2011).
33. Uyanga, I. J. & Idem, R. O. Studies of SO_2- and O_2-induced degradation of aqueous MEA during CO_2 capture from power plant flue gas streams. *Ind. Eng. Chem. Res.* **46**, 2558–2566 (2007).
34. Lee, J.-Y., Keener, T. C. & Yang, Y. J. Potential flue gas impurities in carbon dioxide streams separated from coal-fired power plants. *J. Air Waste Manag. Assoc.* **59**, 725–732 (2009).
35. Gabardo, C. M. et al. Continuous carbon dioxide electroreduction to concentrated multi-carbon products using a membrane electrode assembly. *Joule* **3**, 2777–2791 (2019).
36. Opperman, H., Kerres, J. & Krieg, H. SO_2 crossover flux of Nafion® and sFS-PBI membranes using a chronocoulometric (CC) monitoring technique. *J. Membr. Sci.* **415–416**, 842–849 (2012).
37. Xiao, H., Cheng, T. & Goddard, W. A. III Atomistic mechanisms underlying selectivities in C_1 and C_2 products from electrochemical reduction of CO on Cu(111). *J. Am. Chem. Soc.* **139**, 130–136 (2017).
38. Montoya, J. H., Shi, C., Chan, K. & Nørskov, J. K. Theoretical insights into a CO dimerization mechanism in CO_2 electroreduction. *J. Phys. Chem. Lett.* **6**, 2032–2037 (2015).
39. Li, L. et al. Hydrophobicity graded gas diffusion layer for stable electrochemical reduction of CO_2. *Angew. Chem. Int. Ed.* **61**, e202208534 (2022).
40. Xing, Z., Hu, L., Ripatti, D. S., Hu, X. & Feng, X. Enhancing carbon dioxide gas-diffusion electrolysis by creating a hydrophobic catalyst microenvironment. *Nat. Commun.* **12**, 136 (2021).
41. Kim, C. et al. Tailored catalyst microenvironments for CO_2 electroreduction to multicarbon products on copper using bilayer ionomer coatings. *Nat. Energy* **6**, 1026–1034 (2021).
42. Kusoglu, A. & Weber, A. Z. New insights into perfluorinated sulfonic-acid ionomers. *Chem. Rev.* **117**, 987–1104 (2017).
43. Ren, X., Myles, T. D., Grew, K. N. & Chiu, W. K. S. Carbon dioxide transport in Nafion 1100 EW membrane and in a direct methanol fuel cell. *J. Electrochem. Soc.* **162**, F1221–F1230 (2015).
44. Dinh, C.-T. et al. CO_2 electroreduction to ethylene via hydroxide-mediated copper catalysis at an abrupt interface. *Science* **360**, 783–787 (2018).
45. Bohra, D., Chaudhry, J. H., Burdyny, T., Pidko, E. A. & Smith, W. A. Modeling the electrical double layer to understand the reaction environment in a CO_2 electrocatalytic system. *Energy Environ. Sci.* **12**, 3380–3389 (2019).
46. Huang, J. E. et al. CO_2 electrolysis to multicarbon products in strong acid. *Science* **372**, 1074–1078 (2021).
47. D'Alessandro, D. M., Smit, B. & Long, J. R. Carbon dioxide capture: prospects for new materials. *Angew. Chem. Int. Ed.* **49**, 6058–6082 (2010).
48. Li, J. et al. Silica-copper catalyst interfaces enable carbon-carbon coupling towards ethylene electrosynthesis. *Nat. Commun.* **12**, 2808 (2021).
49. Harmon, N. J. & Wang, H. Electrochemical CO_2 reduction in the presence of impurities: influences and mitigation strategies. *Angew. Chem. Int. Ed.* **61**, e202213782 (2022).
50. Ko, B. H. et al. The impact of nitrogen oxides on electrochemical carbon dioxide reduction. *Nat. Commun.* **11**, 5856 (2020).
51. Xu, Y. et al. Oxygen-tolerant electroproduction of C_2 products from simulated flue gas. *Energy Environ. Sci.* **13**, 554–561 (2020).
52. Kresse, G. & Furthmüller, J. Efficiency of ab-initio total energy calculations for metals and semiconductors using a plane-wave basis set. *Comput. Mater. Sci.* **6**, 15–50 (1996).
53. Kresse, G. & Hafner, J. *Ab initio* molecular-dynamics simulation of the liquid-metal–amorphous-semiconductor transition in germanium. *Phys. Rev. B* **49**, 14251–14269 (1994).
54. Perdew, J. P., Burke, K. & Ernzerhof, M. Generalized gradient approximation made simple. *Phys. Rev. Lett.* **77**, 3865–3868 (1996).
55. Blöchl, P. E. Projector augmented-wave method. *Phys. Rev. B* **50**, 17953–17979 (1994).
56. Grimme, S., Antony, J., Ehrlich, S. & Krieg, H. A consistent and accurate *ab initio* parametrization of density functional dispersion

correction (DFT-D) for the 94 elements H-Pu. *J. Chem. Phys.* **132**, 154104 (2010).
57. Monkhorst, H. J. & Pack, J. D. Special points for Brillouin-zone integrations. *Phys. Rev. B* **13**, 5188–5192 (1976).
58. Methfessel, M. & Paxton, A. High-precision sampling for Brillouin-zone integration in metals. *Phys. Rev. B* **40**, 3616–3621 (1989).

Acknowledgements

This work was financially supported by the National Key Research and Development Program of China (2022YFA1505100 and 2023YFA1507500 to J.L.), the National Natural Science Foundation of China (grant number BE3250011 to J.L.), the Fundamental Research Funds for the Central Universities (23×010301599 to J.L.), Shanghai Pilot Program for Basic Research, Shanghai Jiao Tong University (21TQ1400227 to J.L.), the Ontario Research Foundation: Research Excellence Program (to D.S.), the Natural Sciences and Engineering Research Council (NSERC) of Canada (to D.S and P.P.) and TOTAL SE (to D.S.). Part or all of the XAS measurements described in this paper were performed at the Soft X-ray Microcharacterization Beamline at the Canadian Light Source, a national research facility of the University of Saskatchewan, which is supported by NSERC, the Canada Foundation for Innovation (CFI), the National Research Council (NRC), the Canadian Institutes of Health Research (CIHR), the Government of Saskatchewan and the University of Saskatchewan. We gratefully acknowledge support from the Canada Research Chairs Program. The computational study is supported by the Marsden Fund Council from Government funding (21-UOA-237 to Z.W.) and Catalyst: Seeding General Grant (22-UOA-031-CGS to Z.W.), managed by Royal Society Te Apārangi. Z.W. and R.L. acknowledge the use of New Zealand eScience Infrastructure (NeSI) high-performance computing facilities, consulting support and/or training services as part of this research.

Author contributions

D.S. and J.L. supervised the project. J.L. and P.P. conceived the idea. J.L. and P.P designed and P.P. and R.K.M. carried out all the electrochemical experiments. H. Liu, X.W., A.O. and C.P.O. contributed to electrode fabrication and assisted with the electrochemical testing. R.L. performed DFT calculations with the supervision of Z.W.; S.L. performed COMSOL Multiphysics simulations and F.L. supported the related discussions. J.L. and P.P. performed XAS and Raman measurements, in which Y.H., M.S. and Q.X. assisted with the XAS testing and R.K.M. and J.E.H. assisted with the Raman testing. M.L. and J.Y.H. performed scanning electron microscopy, TEM and X-ray photoemission spectroscopy characterizations, with the contributions of Y.W., Q.Z. and P.L. for data analysis. A.S.Z., B.K. and K.G. assisted with the contact angle measurements. N.S. and Y.C.X. contributed to figure draughting. J.L. wrote the manuscript. D.S., E.H.S., H. Liang and P.P. contributed to manuscript editing. All authors discussed the results and assisted during the manuscript preparation.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41560-024-01577-9.

Correspondence and requests for materials should be addressed to Ziyun Wang, Jun Li or David Sinton.

Peer review information *Nature Energy* thanks Feng Jiao and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature Limited 2024

[1]Department of Mechanical and Industrial Engineering, University of Toronto, Toronto, Ontario, Canada. [2]School of Chemical Sciences, University of Auckland, Auckland, New Zealand. [3]Frontiers Science Center for Transformative Molecules, School of Chemistry and Chemical Engineering, Zhangjiang Institute for Advanced Study, Shanghai Jiao Tong University, Shanghai, China. [4]School of Materials Science and Engineering, Tianjin University, Tianjin, China. [5]Sinopec Shanghai Research Institute of Petrochemical Technology, Shanghai, China. [6]Canadian Light Source, Inc., University of Saskatchewan, Saskatoon, Saskatchewan, Canada. [7]Department of Materials Science and Engineering, University of Toronto, Toronto, Ontario, Canada. [8]Department of Electrical and Computer Engineering, University of Toronto, Toronto, Ontario, Canada. [9]These authors contributed equally: Panagiotis Papangelakis, Rui Kai Miao, Ruihu Lu, Hanqi Liu, Xi Wang. ✉e-mail: ziyun.wang@auckland.ac.nz; lijun001@sjtu.edu.cn; sinton@mie.utoronto.ca